

COMPOSE: COMPOSITIONAL OPERATOR-SCHEDULED EDITING FOR MOLECULAR GENERATION

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Paper under double-blind review

ABSTRACT

Molecular graph generators commonly make one of three compromises: they fix the atom count before denoising, traverse disconnected or valence-violating intermediate graphs, or construct molecules through an irreversible growth order. We introduce COMPOSE, a trans-dimensional generative process defined directly on a bounded space of connected, valence-admissible molecular graphs. Its state transitions form a reversible chemical edit instruction set—atom insertion and deletion, atom substitution, bond-order modification, and ring opening or closure—with a legality predicate evaluated before every commit. A high-entropy valid-graph prior is made stationary under a Metropolis-corrected continuous-time Markov chain (CTMC), while a learned hierarchical scheduler estimates the reverse instruction rates. We prove pathwise constraint preservation, prior invariance, and irreducibility under explicit operator assumptions, and we separate the exact single-issue process from an optional conflict-aware multiple-issue sampler. This construction supports unconditional de novo generation when the initial graph is drawn independently from the prior, and source-conditioned editing when a molecule is supplied by the user.

1 INTRODUCTION

A molecular generator should place probability mass on realistic molecules, but the path used to reach those molecules also matters. Intermediate states are exposed to property guidance, scaffold constraints, human intervention, and iterative evaluation. If that path routinely leaves the molecular domain, a model must learn through states on which chemical features are undefined or misleading, and downstream guidance can act on disconnected or valence-inadmissible graphs. Endpoint sanitization does not resolve this mismatch.

Graph diffusion and flow models have substantially improved molecular distribution learning (Jo et al., 2022; Vignac et al., 2023; Jo et al., 2024; Xu et al., 2024; Siraudin et al., 2025; Qin et al., 2025). Most, however, represent a molecule in a fixed-size tensor after sampling the number of atoms outside the generative trajectory. Methods that change size during generation have begun to appear (Ninniri et al., 2025; Havasi et al., 2025; Franke et al., 2026), but flexible size does not by itself ensure that every state is connected and chemically admissible. Conversely, constrained methods can preserve selected graph properties (Madeira et al., 2024) or preserve molecular validity within a fixed formula and degree sequence (Ruiz-Botella et al., 2026), but these restrictions prevent movement between broad composition and size classes.

We ask whether de novo molecular generation can instead be formulated as stochastic execution of a small, reversible set of chemically closed graph operators. This view is inspired by an instruction-set architecture (ISA), but the analogy is useful only because it induces concrete objects: the molecular graph is architectural state; an instruction is an operator together with atom or bond operands; the legality unit detects chemical and connectivity hazards; the scheduler assigns time- and state-dependent issue rates; and an instruction commits only when its output lies in the declared molecular graph space. The ISA is symmetric in support, while its stochastic rates remain directional. Thus reversibility does not force generation to insert and delete equally often.

The resulting method, COMPOSE, defines a CTMC on the edit graph of connected, valence-admissible molecules. Its forward corruption is not an arbitrary random walk. A prescribed high-entropy prior π_0 is made stationary with a continuous-time Metropolis correction (Hastings, 1970). This prevents

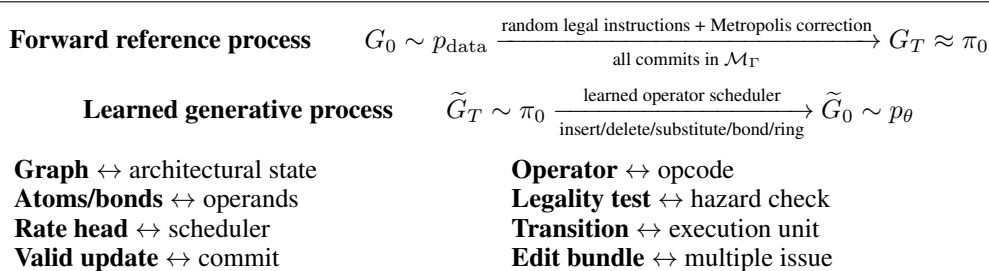


Figure 1: **COMPOSE in one view.** The forward and reverse processes use the same legal edit support but different rates. The computer-architecture framing is operational rather than decorative: it determines the factorization of the generator and the semantics of validity and parallel execution.

the implicit terminal distribution from being dominated by states that merely have many legal edits. The forward process begins at p_{data} , so it carries a nonzero probability current toward π_0 even though each edit has an inverse. The exact time-reversed process is a prior-reversible reference generator tilted by a discrete density ratio. A graph network learns this tilt through a hierarchical operator scheduler.

When generation begins from $G_T \sim \pi_0$ independently of any requested molecule, COMPOSE is an unconditional de novo generator. When G_T is instead supplied by a user, the same transition system supports molecule-to-molecule editing. A valid graph prior is therefore structured noise, not automatically a conditioning molecule.

Our contributions are:

- **A trans-dimensional chemical edit process.** We formalize a reversible operator family whose committed transitions remain connected and valence-admissible while changing atom count, composition, bond order, and cycle structure.
- **A controlled valid-graph prior.** We define a coarse-statistic Gibbs prior and a Metropolis-corrected forward CTMC for which that prior is invariant, rather than accepting the unknown degree bias of uniformly sampled legal actions.
- **Operator-scheduled reverse dynamics.** We factor reverse rates into an event clock, an operator-family scheduler, and operand-specific heads, and train them from reversed forward trajectories without requiring a closed-form corruption marginal.
- **Theory and diagnostics beyond endpoint validity.** We establish closure, detailed balance, and reachability under stated assumptions, and propose tests for mixing, source erasure, operation utilization, path validity, and sampling approximation.

2 BACKGROUND AND POSITIONING

Diffusion and flow on graphs. Continuous-state graph score models perturb adjacency and node features through SDEs (Jo et al., 2022). Discrete graph diffusion instead uses categorical node and edge transitions (Vignac et al., 2023); related continuous-time models offer flexible quality–cost trade-offs (Campbell et al., 2022; Xu et al., 2024; Siraudin et al., 2025). Mixture, flow-matching, and belief-space formulations improve topology modeling, probability paths, or sampling efficiency (Jo et al., 2024; Qin et al., 2025; Jiang et al., 2026; Petersen et al., 2026). Recent molecule-specific systems additionally exploit fragment hierarchies or chemically structured categorical states (Lee et al., 2026; Jung et al., 2026). These methods are strong distribution learners, but they do not define a reversible trans-dimensional atomic edit process whose entire committed trajectory is restricted to connected, valence-admissible graphs.

Hard constraints. ConStruct projects edge-denoising trajectories onto classes satisfying selected edge-deletion-invariant graph constraints (Madeira et al., 2024). CoCoGraph uses connectedness-checked double-edge swaps; the operation preserves each atom’s degree and therefore stays within a

Table 1: Positioning by generative capability. A check mark denotes support-level behavior of the committed trajectory, not an endpoint statistic. "Source init." means that a user-specified molecule can initialize the demonstrated generative procedure. The table is a scope comparison, not a quality ranking.

Method	Variable N	Bidirectional size edits	Valence by support	Connected by support	De novo	Source init.
GDSS (Jo et al., 2022)	✗	✗	✗	✗	✓	✗
DiGress (Vignac et al., 2023)	✗	✗	✗	✗	✓	✗
DeFoG (Qin et al., 2025)	✗ [†]	✗	✗	✗	✓	✗
ConStruct (Madeira et al., 2024)	✗	✗	$\Delta^{\ddagger}$	$\Delta^{\ddagger}$	✓	✗
CoCoGraph (Ruiz-Botella et al., 2026)	✗	✗	✓	✓	✓ [§]	✗
GraphXForm (Pirnay et al., 2025)	✓	✗	✓	✓	✓	✓
GrIDDD (Ninniri et al., 2025)	✓	$\Delta^{\S}$	✗	✗	✓	✓
Morph (Franke et al., 2026)	✓	✓	✗	✗	✓	✓
COMPOSE	✓	✓	✓	✓	✓	✓

[†]The graph path is fixed-cardinality after node count is supplied or sampled. [‡]ConStruct enforces user-selected graph constraints; it does not instantiate the joint molecular valence-and-connectedness grammar used here. [§]CoCoGraph samples within a supplied or sampled formula/degree-sequence fiber. ^{||}GraphXForm can initialize from an existing structure but modifies it through additions rather than reversible size edits. [¶]GrIDDD uses monotone deletion in corruption and insertion in generation rather than freely bidirectional corrections within a trajectory.

fixed formula and degree-sequence fiber while converging toward a maximum-entropy graph distribution in that fiber (Ruiz-Botella et al., 2026). This is the closest precedent for valid intermediates, but it cannot change molecular size, elemental composition, degree sequence, or total valence allocation.

Variable-size and edit-based generation. GraphArm removes nodes in an autoregressive diffusion and reconstructs them in a learned order (Kong et al., 2023). GrIDDD introduces monotone node deletion and insertion for size-adaptive graph diffusion (Ninniri et al., 2025). Edit Flows formulates insertion, deletion, and substitution as a sequence-space CTMC (Havasi et al., 2025), and Morph extends flexible-size jump-flow modeling to geometric molecular graphs with unbalanced optimal transport (Franke et al., 2026). GraphXForm uses masked atom and bond additions for valid sequential construction (Pirnay et al., 2025). COMPOSE differs in the conjunction of bidirectional trans-dimensional edits, hard connectivity and valence support, an explicit target-independent prior, and de novo distribution learning.

3 PROBLEM SETUP

Molecular graph grammar. Let $G = (V, E, \ell, b)$ be a finite undirected graph. The node label ℓ_v contains an element, formal charge, and any discrete valence state represented by the implementation; $b_{uv} \in \mathcal{B}$ is a bond type, with $b_{uv} = 0$ denoting no edge. A graph grammar Γ specifies the allowed labels, bond types, implicit-hydrogen convention, and admissible incident bond-order sums. We write

$$\mathcal{M}_{\Gamma} = \{G : 1 \leq |V| \leq N_{\max}, G \text{ is connected, and } G \text{ is valence-admissible under } \Gamma\}. \quad (1)$$

The bound $N_{\max}$ makes $\mathcal{M}_{\Gamma}$ finite. Our guarantees concern this declared graph grammar, not all bonding phenomena: delocalized, multicenter, radical, and organometallic chemistry require explicit representation if they are to be included. We recommend a Kekulized heavy-atom graph with implicit hydrogens for the first implementation and an independent RDKit sanitization audit (Landrum & Contributors, 2026).

Unconditional versus conditional generation. A generation run is unconditional when its initial state is sampled independently,

$$G_T \sim \pi_0, \quad G_T \perp G_{\text{requested}}, \quad (2)$$

and no molecule-specific scaffold is supplied. A hierarchical prior $c \sim p_0(c)$, $G_T \sim \pi_0(G | c)$ is still unconditional after marginalizing c ; fixing c yields attribute-conditioned de novo generation. Setting $G_T = G_{\text{source}}$ yields source-molecule editing.

Table 2: Minimal COMPOSE ISA. More expressive insert/delete operators may be substituted, provided closure and inverse semantics are tested. A ring operation can be a primitive bond edit or an atomic macro compiled from valid micro-operations.

Family	Forward instruction	Inverse	Commit condition (in addition to local valence)
Atom birth/death	insert labeled leaf at u	delete that leaf	$ V < N_{\max}$; deletion leaves at least one atom
Atom substitution	$\ell_u \leftarrow \ell'_u$	restore ℓ_u	incident bond order admissible under new label
Bond modification	$b_{uv} \leftarrow b_{uv} \pm 1$	opposite change	zero-order deletion allowed only if connectivity remains
Ring close/open	add bond (u, v)	delete bond (u, v)	opening edge is not a bridge; closure has spare capacity
Optional macro	compiled motif/ring edit	inverse program	every exposed microstate valid, or atomic commit semantics

4 COMPOSE

4.1 A REVERSIBLE CHEMICAL EDIT ISA

An instruction is $a = (k, o, \xi)$, where $k \in \mathcal{K}$ is an operator family, o is a tuple of atom or bond operands, and ξ contains labels or bond parameters. The deterministic executor T_a is guarded by a legality predicate

$$\text{Legal}(G, a) = 1 \implies T_a(G) \in \mathcal{M}_\Gamma. \quad (3)$$

Every committed instruction has an inverse $\bar{a}$ satisfying $T_{\bar{a}}(T_a(G)) = G$. The concrete operand schemas depend on the user’s implementation; Table 2 gives a minimal set sufficient for the theory.

The symmetric support is not a symmetric schedule. A learned reverse process may be insertion-heavy early and deletion-heavy only for correction. Symmetry provides legal retraction, an undirected edit geometry, and a tractable reverse proposal probability.

4.2 A HIGH-ENTROPY VALID-GRAPH PRIOR

A single atom, a fixed scaffold, or an all-tree source imposes a strong construction order. A prior composed of real training molecules instead risks identity transport and analogue generation. We use a Gibbs distribution on the valid state space,

$$\pi_0(G) = \frac{1}{Z_\eta} \mathbf{1}\{G \in \mathcal{M}_\Gamma\} \exp[-U_\eta(f(G))], \quad (4)$$

where $f(G)$ contains only coarse statistics, for example

$$f(G) = (N, \mathbf{n}_{\text{element}}, Q_{\text{formal}}, B_{\text{tot}}, \rho, h_{\text{valence}}), \quad \rho = |E| - |V| + 1. \quad (5)$$

The parameters η may be chosen by maximum-entropy moment matching to smoothed training-set marginals. Crucially, U_η excludes fingerprints, scaffolds, pharmacophores, exact fragment frequencies, and nearest-neighbor terms. The prior overlaps the data in scale and chemical support while destroying high-order drug-like organization.

We initialize prior chains with a constructive sampler: sample size and composition, build a random valence-feasible spanning tree, add random legal closure bonds, and randomize bond orders. The constructor need not itself have distribution (4); under the irreducibility conditions below, a sufficiently long prior chain approaches the evaluable target. Finite-run samples are approximate, so convergence and initialization sensitivity are measured rather than assumed.

4.3 METROPOLIS-CORRECTED FORWARD CTMC

Let $c_t(G, G')$ be a fixed proposal rate obtained by aggregating all legal instruction aliases,

$$c_t(G, G') = \sum_{a: T_a(G)=G'} \beta_{k(a)}(t) \rho(a | G, k(a)), \quad (6)$$

where $\beta_k(t)$ schedules an operator family and ρ selects operands and parameters. The continuous-time Metropolis generator is

$$Q_t(G, G') = c_t(G, G') \min\left\{1, \frac{\pi_0(G') c_t(G', G)}{\pi_0(G) c_t(G, G')}\right\}, \quad G' \neq G, \quad (7)$$

with $Q_t(G, G) = -\sum_{G' \neq G} Q_t(G, G')$. The partition function in (4) cancels. The inverse direction must have nonzero proposal support whenever the forward direction does; accordingly, operator schedules are paired for birth/death, bond increase/decrease, and ring close/open.

Unlike a discrete propose/reject implementation, (7) directly defines accepted jump rates. A Gillespie sampler draws the waiting time from the total exit rate and then selects a committed instruction (Gillespie, 1977). The correction is used in the *forward* process because π_0 is known. Exact MH correction toward the unknown molecular data density would not be available in reverse.

Forward trajectories start at $X_0 \sim p_{\text{data}}$. Their edge current is

$$J_t(G, G') = p_t(G)Q_t(G, G') - p_t(G')Q_t(G', G). \quad (8)$$

Detailed balance holds only when $p_t = \pi_0$; thus a symmetric ISA does not eliminate the noising current.

4.4 REVERSE OPERATOR SCHEDULING

Let $Y_s = X_{T-s}$ and let p_t denote the forward marginal. The exact reverse rate is

$$R_s^*(G, G') = Q_{T-s}(G', G) \frac{p_{T-s}(G')}{p_{T-s}(G)}. \quad (9)$$

Because (7) is reversible with respect to π_0 , define the relative log-density $h_t(G) = \log p_t(G) - \log \pi_0(G)$ to obtain

$$R_s^*(G, G') = Q_{T-s}(G, G') \exp[h_{T-s}(G') - h_{T-s}(G)]. \quad (10)$$

This is a discrete score along ISA edges. We parameterize a nonnegative marked-action rate and factor it hierarchically,

$$R_{\theta,s}(a | G) = \lambda_{\theta}(G, s) p_{\theta}(k(a) | G, s) p_{\theta}(o, \xi | k(a), G, s), \quad (11)$$

where the family distribution is normalized over families with at least one legal action and each operand distribution is normalized over that family’s legal parameterizations. The first head predicts the event clock, the second is the operator scheduler, and family-specific heads select operands. Consequently,

$$\Lambda_{\theta}(G, s) = \sum_{a \in \mathcal{A}(G)} R_{\theta,s}(a | G) = \lambda_{\theta}(G, s), \quad (12)$$

so the exact exit rate does not require summing a large molecular neighborhood. State-transition rates aggregate aliases,

$$R_{\theta,s}(G, G') = \sum_{a: \tau_a(G)=G'} R_{\theta,s}(a | G). \quad (13)$$

The factorization prevents an operator with many parameterizations from dominating a flat action softmax merely by cardinality. An optional energy-consistent parameterization sets the log-rate correction to $h_{\theta}(\tau_a G, t) - h_{\theta}(G, t)$; the direct action-rate model is more flexible but can be regularized for cycle consistency.

Trajectory likelihood. A simulated forward trajectory, reversed in time, is an exact sample from the reverse path law. Let its reverse jumps occur at $0 < s_1 < \dots < s_m < T$. Up to a constant independent of θ , the negative state-path log-likelihood is

$$\mathcal{L}_{\text{path}}(\theta) = \mathbb{E} \left[\int_0^T \lambda_{\theta}(Y_s, s) ds - \sum_{j=1}^m \log \sum_{a: \tau_a(Y_{s_j}^-) = Y_{s_j}} R_{\theta,s_j}(a | Y_{s_j}^-) \right]. \quad (14)$$

With a canonical one-to-one action encoding, each inner sum has one term; otherwise the log-sum prevents action aliases from changing the learned state process. The holding-time integral is estimated by quadrature or unbiased time sampling within each constant-state interval. This objective requires neither a closed-form $q_t(G_t | G_0)$ nor sampled-neighbor exit-rate estimates. Alternative unnormalized edge-rate parameterizations do require a neighborhood estimator or uniformization. Generator Matching provides another estimator for arbitrary Markov generators (Holderrieth et al., 2025).

4.5 OPTIONAL CONFLICT-AWARE MULTIPLE ISSUE

The exact stochastic process is single-jump: simultaneous jumps have probability zero in a CTMC. We therefore treat multiple issue as a sampling accelerator, not as the theoretical definition of COMPOSE. At one network evaluation, the model proposes candidate instructions and a conflict-aware scheduler selects a bundle B . A bundle commits only when there exists a canonical sequential order in which every intermediate graph remains in $\mathcal{M}_\Gamma$ and the resulting state is independent of the permitted order. Pairwise conflicts capture shared operands and read/write hazards, but valence capacity and joint connectivity can require a hypergraph or integer constraint. We call this variant COMPOSE-MI; "maximum independent set" terminology is used only if that optimization is literally solved.

5 THEORETICAL PROPERTIES

We state the principal results here and give proofs in Section B. They are conditional on the declared grammar and operator implementation; this makes the scope auditable.

Proposition 1 (Pathwise chemical closure). *If $G_0 \in \mathcal{M}_\Gamma$ and every committed action satisfies (3), then every state visited by the forward CTMC, the learned reverse CTMC, and any serializable multi-issue bundle lies in $\mathcal{M}_\Gamma$ almost surely.*

Proposition 2 (Prior reversibility). *For each t , the generator in (7) satisfies*

$$\pi_0(G)Q_t(G, G') = \pi_0(G')Q_t(G', G). \quad (15)$$

Therefore $\pi_0 Q_t = 0$ and π_0 is invariant. The claim remains true for a time-dependent operator schedule if every active transition has inverse support.

Assumption 3 (Canonical primitive coverage). The ISA can (i) delete any non-bridge edge, (ii) delete any leaf other than the sole atom, (iii) insert a labeled leaf with any target-legal bond order, (iv) add any target-legal edge between existing atoms, and (v) substitute the isolated seed label. Every primitive and inverse receives positive proposal rate on its legal domain.

Theorem 4 (Reachability). *Under Assumption 3, the undirected edit graph induced on $\mathcal{M}_\Gamma$ is connected.*

Corollary 5 (Irreducibility and convergence). *Let $\bar{Q}$ be a time-homogeneous instance of (7) with positive rate on every legal edit edge. Then the finite-state CTMC is irreducible, π_0 is its unique stationary distribution, and $p_t \rightarrow \pi_0$ as $t \rightarrow \infty$. The same conclusion holds for a scalar schedule $Q_t = \gamma(t)\bar{Q}$ whenever $\gamma(t) > 0$ and $\int_0^\infty \gamma(t) dt = \infty$.*

The constructive proof opens non-tree edges, repeatedly deletes leaves to reach one atom, and reverses the procedure to build any target spanning tree before adding its remaining edges. The argument also exposes precisely which operator restrictions would partition chemical space. A finite training horizon still produces only an approximation to π_0 , which is why terminal mixing is measured explicitly.

Proposition 6 (Endpoint error decomposition). *Let K^* be the exact reverse transition kernel from T to 0, and let K_θ be the learned reverse kernel. If P^* and P_θ^{pT} are the corresponding reverse path laws initialized at the true terminal marginal p_T , then*

$$\text{TV}(\pi_0 K_\theta, p_{\text{data}}) \leq \text{TV}(\pi_0, p_T) + \sqrt{\frac{1}{2} D_{\text{KL}}(P^* \| P_\theta^{pT})}. \quad (16)$$

Thus generation error separates into terminal mixing error and learned reverse-path error. Objective (14) minimizes the latter up to a constant.

These results do not claim that graph-valid states are synthetically accessible or pharmacologically useful. They establish support, stationarity, and learnability properties for the declared representation.

6 EXPERIMENTAL PROTOCOL

The experimental design is organized around six questions.

RQ1 Does COMPOSE match the molecular data distribution while retaining diversity and nontrivial novelty?

- RQ2** Are all committed intermediate states connected and valence-admissible without repair?
- RQ3** Does the forward chain erase source identity and converge to the declared prior?
- RQ4** Does Metropolis correction remove measurable action-degree or size bias relative to uncorrected legal edits?
- RQ5** Does hierarchical operator scheduling improve learning over a flat action head or fixed family schedule?
- RQ6** What speed–distribution trade-off is introduced by conflict-aware multiple issue?

Data and representation. GuacaMol is the primary large-scale unconditional generation benchmark (Brown et al., 2019). MOSES (Polykovskiy et al., 2020) provides a second drug-like distribution benchmark, and QM9 provides a small-molecule stress test in which exhaustive operator audits and size-extrapolation experiments are tractable. For each dataset we use the official split when available, deduplicate by the frozen canonical representation, and report the exact RDKit version, salt/fragment handling, element and charge vocabulary, stereochemistry policy, Kekulization failures, and counts after every filter. All methods in a matched comparison receive the same processed split.

Baselines and fairness. We compare against JTVAE (Jin et al., 2018), GDSS, DiGress, GruM, DISCO, Cometh, DeFoG, ConStruct, CoCoGraph, and GrIDDD. GraphXForm and Morph are included in capability and conditional-editing comparisons where their task and representation align. We separate (i) a contextual table of numbers reported by original papers from (ii) a matched evaluation using the same data split, sample count, sanitization code, and hardware. Only the matched track supports statements of superiority. A baseline that cannot be faithfully retrained is labeled rather than silently mixed with matched results.

Endpoint quality. We report RDKit sanitization validity, uniqueness, exact train-set novelty, GuacaMol KL score, and either raw Fréchet ChemNet distance (lower is better) or GuacaMol’s transformed FCD score (higher is better), never both under the same label (Preuer et al., 2018). Exact novelty is supplemented by nearest-training-set fingerprint similarity, Bemis–Murcko scaffold novelty (Bemis & Murcko, 1996), maximum-common-subgraph similarity, duplicate rate, and property-distribution distances. We do not equate sanitization with synthesizability.

Trajectory quality. For every sampled path we record the fraction of committed states satisfying the internal grammar, RDKit sanitization, and single-component connectivity; operator proposal, legality, acceptance, and execution rates; edit count; and the maximum excursion in size, composition, and cycle rank. These measurements verify the implementation even when internal validity is guaranteed by construction.

Mixing and source erasure. We compare terminal corrupted samples $X_T \mid X_0$ with independent prior samples $G \sim \pi_0$. Diagnostics include a two-sample classifier AUC, kernel MMD over coarse and fingerprint features, initial–terminal atom survival, maximum common substructure, scaffold retention, and correlations between initial and terminal molecular properties. Chains are initialized from structurally distinct source strata to expose slow-mixing classes. A terminal classifier near chance and low source retention are necessary for a credible unconditional prior.

Statistics and compute. All learned models use at least three independent seeds. We report mean, standard deviation, and paired bootstrap confidence intervals over a common generated sample set. Throughput is measured after warm-up on identical hardware, with batch size, neural function evaluations, accepted instructions, and wall-clock time stated separately. Hyperparameter selection uses validation metrics fixed before test evaluation.

Confirmatory status of developmental diagnostics. Pilot observations used to choose the prior, operator set, or schedule are not treated as final evidence. Every pilot-derived hypothesis—including single-atom, tree-only, tree-plus-ring, and uncorrected-walk behavior—is rerun with the frozen preprocessing, architecture budget, seed policy, and evaluator. Design diagnostics that remain informative but are not needed for a main claim are reported in Section F.8; irreproducible or incomparable pilot numbers are omitted.

Table 3: Matched GuacaMol distribution learning. “All-step valid” is the fraction of committed trajectory states satisfying the declared internal grammar. FCD denotes raw Fréchet ChemNet distance (lower is better).

Method	Valid $\uparrow$	Unique $\uparrow$	Novel $\uparrow$	FCD $\downarrow$	KL score $\uparrow$	All-step valid $\uparrow$	Samples/s $\uparrow$
BWFlow	TBD	TBD	TBD	TBD	TBD	TBD	TBD
GraphBSI	TBD	TBD	TBD	TBD	TBD	–	TBD
DiGress	TBD	TBD	TBD	TBD	TBD	TBD	TBD
GruM	TBD	TBD	TBD	TBD	TBD	TBD	TBD
DISCO	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Cometh	TBD	TBD	TBD	TBD	TBD	TBD	TBD
DeFoG	TBD	TBD	TBD	TBD	TBD	TBD	TBD
ConStruct	TBD	TBD	TBD	TBD	TBD	TBD	TBD
CoCoGraph	TBD	TBD	TBD	TBD	TBD	TBD	TBD
GrIddd	TBD	TBD	TBD	TBD	TBD	TBD	TBD
COMPOSE	TBD	TBD	TBD	TBD	TBD	TBD	TBD

Table 4: Core ablations. The acceptance and source-erasure columns prevent endpoint metrics from hiding a failed prior.

Variant	FCD $\downarrow$	All-step valid $\uparrow$	Mean accept. $\uparrow$	Terminal classifier AUC $\rightarrow .5$	Atom survival $\downarrow$
Single-atom prior	TBD	TBD	TBD	TBD	TBD
Random-tree prior	TBD	TBD	TBD	TBD	TBD
Coarse Gibbs prior, no MH	TBD	TBD	TBD	TBD	TBD
Gibbs prior + MH, flat head	TBD	TBD	TBD	TBD	TBD
COMPOSE, single issue	TBD	TBD	TBD	TBD	TBD
COMPOSE-MI	TBD	TBD	TBD	TBD	TBD

7 RESULTS AND ANALYSIS

7.1 UNCONDITIONAL DISTRIBUTION LEARNING

On GuacaMol, COMPOSE obtains validity **TBD**, uniqueness **TBD**, exact train-set novelty **TBD**, raw FCD **TBD**, and KL score **TBD**. The strongest matched baseline, **TBD**, obtains validity **TBD** and FCD **TBD** under the same sample count and evaluator. On the secondary datasets, the corresponding distribution and graph statistics are reported in Table 10. Together with scaffold novelty and nearest-training-neighbor similarity in Table 8, these results distinguish distributional fidelity from merely avoiding exact duplicates.

7.2 VALIDITY ALONG THE ENTIRE PATH

Across all generated trajectories, **TBD** of committed states satisfy the internal grammar and **TBD** remain connected. Endpoint validity therefore does not hide invalid intermediate graphs. Independent sanitization and failure categories are reported separately because the formal guarantee is exactly “connected and valence-admissible under Γ ,” not universal chemical correctness.

7.3 PRIOR MIXING AND THE ROLE OF MH

The terminal two-sample classifier has AUC **TBD**, while initial atoms survive with rate **TBD** and initial-terminal MCS ratio is **TBD**. The Metropolis correction changes the terminal prior discrepancy from **TBD** without correction to **TBD**, with mean acceptance **TBD**. These measurements test both prerequisites of unconditional generation: convergence toward the declared prior and erasure of the particular source molecule.

7.4 SCHEDULER AND MULTIPLE-ISSUE ABLATIONS

Hierarchical scheduling changes FCD from **TBD** for a flat action head to **TBD**, while exposing interpretable family-wise rate profiles. COMPOSE-MI reduces neural evaluations from **TBD** to **TBD**

and yields speedup **TBD** at distribution gap **TBD** relative to exact single issue; it is therefore reported as an approximation rather than as the defining process.

8 LIMITATIONS AND CONCLUSION

Limitations. The guarantee is scoped to the declared graph grammar and does not automatically cover resonance, tautomerism, stereochemistry, organometallics, multicenter bonding, synthesizability, or biological utility. Irreducibility does not imply rapid finite-time mixing; prior equilibration and forward simulation may be costly. Multiple issue remains an approximation unless modeled as a compound-jump process, and training on GuacaMol targets its distribution rather than chemistry far outside it.

Conclusion. COMPOSE defines molecular generation as stochastic execution on a connected, valence-admissible edit graph: a reversible ISA supplies legal moves, a Metropolis-corrected reference process supplies a known high-entropy prior, and a learned scheduler supplies the reverse current. Unconditional generation and source editing are thus two initializations of one process. The empirical burden is explicit: quality, source erasure, path validity, mixing, and the quality–cost effect of scheduling.

Reproducibility statement. The anonymous supplement specifies the grammar, proofs, prior energy, alias-aware proposal rates, trajectory simulator, estimator, preprocessing, seeds, evaluation, and compute accounting; all tables are generated from frozen machine-readable outputs.

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A EXACT SCOPE OF THE MOLECULAR GRAPH GRAMMAR

We make the guarantee machine-checkable by specifying the exact grammar in Table 5 and exporting the same specification as code.

Table 5: Frozen molecular grammar and preprocessing specification.

Item	Exact specification
Allowed elements	TBD
Formal charges and valence states	TBD
Bond types	TBD
Implicit/explicit hydrogen policy	TBD
Aromaticity/Kekulization policy	TBD
Stereochemistry policy	TBD
Minimum/maximum heavy atoms	TBD
Salt and multi-fragment policy	TBD
RDKit version and sanitization flags	TBD
Canonicalization and duplicate key	TBD

A.1 OPERATOR CONTRACTS

For every operator family, the implementation exposes four executable functions:

$$\text{enumerate_legal}(G, k), \quad \text{apply}(G, a), \quad (17)$$

$$\text{inverse}(G, a), \quad \text{proposal_logprob}(G, a, t). \quad (18)$$

We exhaustively enumerate all graphs up to a small size to test closure, inverse recovery, proposal aggregation, permutation equivariance, and agreement with the independent sanitizer. Property-based tests cover larger graphs and adversarial ring and articulation cases.

Table 6: Implementation-level ISA specification.

Opcode	Operand schema	State update	Exact legality predicate	Inverse alias handling
INSERT	TBD	TBD	TBD	TBD
DELETE	TBD	TBD	TBD	TBD
SUBSTITUTE	TBD	TBD	TBD	TBD
BOND-MODIFY	TBD	TBD	TBD	TBD
RING-INSTALL/OPEN	TBD	TBD	TBD	TBD

Ring macros. If a ring installation is compiled into primitive instructions $a_{1:m}$, state whether the microstates are exposed to the stochastic process. If exposed, require $G_i = T_{a_i} \circ \dots \circ T_{a_1}(G) \in \mathcal{M}_\Gamma$ for every i . If only the final state is guaranteed, execute the macro as one atomic compound jump and count it as a primitive in the generator. Mixing these semantics would invalidate an “all intermediate states” claim.

B PROOFS

B.1 PROOF OF PATHWISE CLOSURE

A pure-jump CTMC changes state only at its countable set of jump times. By hypothesis, if the pre-jump state $G \in \mathcal{M}_\Gamma$, any committed action maps it to $T_a(G) \in \mathcal{M}_\Gamma$. Induction over jump count proves that every visited state lies in $\mathcal{M}_\Gamma$. The same argument applies to the learned reverse process because its support is masked to the same legal actions. For a multiple-issue bundle, serializability supplies an ordering of constituent instructions for which every partial commit lies in $\mathcal{M}_\Gamma$; applying the single-instruction argument to that order proves the result. $\square$

B.2 PROOF OF PRIOR INVARIANCE

For $G \neq G'$, set

$$r = \frac{\pi_0(G')c_t(G', G)}{\pi_0(G)c_t(G, G')}. \quad (19)$$

Then

$$\pi_0(G)Q_t(G, G') = \pi_0(G)c_t(G, G') \min\{1, r\} \quad (20)$$

$$= \min\{\pi_0(G)c_t(G, G'), \pi_0(G')c_t(G', G)\} \quad (21)$$

$$= \pi_0(G')Q_t(G', G). \quad (22)$$

Summing over G and using the diagonal definition gives $\pi_0 Q_t = 0$. A time-inhomogeneous chain preserves π_0 because this identity holds at every time. $\square$

B.3 PROOF OF REACHABILITY

Take arbitrary $G, H \in \mathcal{M}_\Gamma$. Choose a spanning tree of G . Every edge outside that tree is non-bridging, so assumption (i) removes those edges while preserving connectivity; removing bonds cannot exceed a local maximum-valence constraint. The remaining graph is a tree. Repeatedly remove a leaf using assumption (ii) until one atom remains, substituting its label to a canonical seed if necessary. This constructs a valid path $G \rightsquigarrow G_\circ$.

Reverse the construction for H . Starting from $G_\circ$, set the seed label and insert the vertices of a chosen spanning tree of H in leaf order, directly assigning their target labels and tree-bond orders. Each partial graph is a connected subgraph of the target and has no greater incident bond-order sum than the target. Finally add every non-tree target edge. Each addition remains within the target valence budget and preserves connectivity. Thus $G_\circ \rightsquigarrow H$, and concatenation gives $G \rightsquigarrow H$. $\square$

B.4 PROOF OF IRREDUCIBILITY AND CONVERGENCE

Connectivity of the edit graph and strictly positive legal-edge rates make $\bar{Q}$ irreducible. Because $\mathcal{M}_\Gamma$ is finite, the CTMC is non-explosive and has a unique invariant distribution. Theorem 2 identifies that distribution as π_0 , and irreducibility of a finite continuous-time chain gives $p_t \rightarrow \pi_0$. For $Q_t = \gamma(t)\bar{Q}$, define operational time $\tau(t) = \int_0^t \gamma(u) du$. The transition operator is $\exp(\tau(t)\bar{Q})$; hence $\tau(t) \rightarrow \infty$ implies convergence to π_0 . $\square$

What can break the proof. The proof fails if deletions cannot remove all leaves, if ring opening cannot remove an arbitrary non-bridge edge, if some labels cannot be inserted or substituted, or if the grammar forbids chemically under-saturated partial graphs even after implicit-hydrogen completion. These are testable design choices rather than minor technicalities.

B.5 PROOF OF THE ENDPOINT-ERROR BOUND

The exact reverse kernel satisfies $p_T K^* = p_{\text{data}}$. By the triangle inequality and contraction of total variation under a Markov kernel,

$$\text{TV}(\pi_0 K_\theta, p_{\text{data}}) \leq \text{TV}(\pi_0 K_\theta, p_T K_\theta) + \text{TV}(p_T K_\theta, p_T K^*) \quad (23)$$

$$\leq \text{TV}(\pi_0, p_T) + \text{TV}(P_\theta^{p_T}, P^*). \quad (24)$$

Pinsker's inequality yields the second term in (16). The expected negative log likelihood in (14) differs from $D_{\text{KL}}(P^* \| P_\theta^{p_T})$ by the entropy of P^* , which is independent of θ . $\square$

B.6 PERMUTATION EQUIVARIANCE

Let σ be a permutation of node indices. If the proposal satisfies $c_t(\sigma G, \sigma G') = c_t(G, G')$, the energy features are permutation invariant, and the neural scheduler is equivariant over operands, then both Q_t and $R_{\theta,t}$ commute with σ . The induced distribution is therefore defined on unlabeled molecular graphs despite implementation on indexed tensors. Action-alias aggregation must also be permutation-consistent.

C PRIOR CONSTRUCTION AND CALIBRATION

C.1 MAXIMUM-ENTROPY INTERPRETATION

Given desired coarse moments $\mu = \mathbb{E}[f(G)]$, the distribution of maximum entropy on $\mathcal{M}_\Gamma$ subject to those moments has exponential-family form (4). The dual objective is

$$\min_{\eta} \log Z_{\eta} + \eta^{\top} \mu, \quad (25)$$

up to the sign convention in U_{η} . Exact gradients require expectations under π_0 ; stochastic approximation with persistent MCMC chains is sufficient for the modest feature set proposed here. A simpler first model may hand-fit separable penalties to broad empirical quantiles and reserve exact moment matching for an ablation.

C.2 PRIOR FEATURE SET

We use the smallest feature set that avoids pathological scale mismatch:

- a smoothed heavy-atom count histogram or truncated negative-binomial energy;
- smoothed global element and charge counts, preferably with a Dirichlet–multinomial interpretation;
- total bond-order mass or a coarse unsaturation statistic;
- cycle rank conditioned weakly on size;
- a coarse histogram of occupied valence states.

Do not fit ring-system identities, functional groups, scaffolds, fingerprints, or fragment co-occurrences. Those high-order correlations are the learning signal.

C.3 CONSTRUCTIVE INITIALIZER

Algorithm 1 Approximate initializer for a prior MCMC chain

- 1: Sample N , element/charge counts, and a cycle-rank budget from smoothed coarse marginals.
 - 2: Allocate valence capacities consistent with the sampled labels; reject infeasible budgets.
 - 3: Draw a random connected tree using only capacity-feasible leaf additions.
 - 4: Add random legal closure bonds until the budget is met or no legal closure remains.
 - 5: Randomize non-aromatic bond orders under the grammar; derive a consistent Kekulized representation.
 - 6: Return G as an initializer, then run the MH-corrected prior chain for burn-in.
-

The initializer is evaluated separately from approximately equilibrated π_0 samples. Otherwise the method may silently learn to denoise constructor artifacts rather than the declared prior.

C.4 PRIOR DIAGNOSTICS

For each feature and several held-out graph statistics, we report trace convergence, effective sample size, and between-chain diagnostics from diverse starts. We record operator-family acceptance and rejected-reason histograms, and compare independently generated prior-sample banks to detect hidden nonstationarity. Since graph MCMC lacks a universal scalar finite-time convergence certificate, the evidence is reported as a collection of diagnostics rather than as a claim of exact mixing.

D TRAINING AND SAMPLING DETAILS

D.1 FORWARD SIMULATION

Full neighborhood enumeration may be costly. Exact alternatives include composition/rejection sampling over operator families, dominating-rate thinning, or uniformization. Whichever estimator is used must preserve the state-to-state rate in (7); sampling one instruction alias and ignoring other aliases can change the stationary distribution.

Algorithm 2 Single-issue forward reference simulation

Require: $G_0 \in \mathcal{M}_\Gamma$, horizon T , prior energy U_η , proposal schedule c_t

- 1: $t \leftarrow 0, G \leftarrow G_0$, trajectory $\omega \leftarrow [(0, G)]$
- 2: **while** $t < T$ **do**
- 3: Construct or sample the legal candidate neighborhood $\mathcal{N}(G)$.
- 4: **for all** $G' \in \mathcal{N}(G)$ **do**
- 5: Compute alias-aggregated $c_t(G, G')$ and $c_t(G', G)$.
- 6: Set $q(G, G')$ by (7).
- 7: $\Lambda \leftarrow \sum_{G' \in \mathcal{N}(G)} q(G, G')$; sample $\Delta \sim \text{Exp}(\Lambda)$.
- 8: **if** $t + \Delta > T$ **then**
- 9: **break**
- 10: Sample G' with probability $q(G, G')/\Lambda$; set $t \leftarrow t + \Delta, G \leftarrow G'$.
- 11: Append (t, G) to ω .
- 12: **return** ω

D.2 REVERSE PATH TRAINING

For each data molecule, simulate a forward path, reverse its states and jump times, and optimize (14). Under (11), the normalized operator and operand distributions give the exact identity

$$\Lambda_\theta(G, s) = \sum_{a \in \mathcal{A}(G)} R_{\theta,s}(a \mid G) = \lambda_\theta(G, s). \quad (26)$$

For a holding interval $[u, v)$ with state G , one unbiased estimator of its event integral is

$$\int_u^v \lambda_\theta(G, s) \, ds = (v - u) \mathbb{E}_{\tau \sim \text{Unif}(u, v)} [\lambda_\theta(G, \tau)], \quad (27)$$

with additional time samples reducing variance. For an observed state transition $G \rightarrow G'$, the jump contribution is the log-sum-exp over all action aliases in (13); a canonical action encoding reduces it to one term. If an alternative model assigns independent unnormalized rates to edit edges, its exit term must instead be computed by exact enumeration, sampled-neighbor estimation, thinning, or uniformization.

Algorithm 3 Reverse trajectory maximum-likelihood training

- 1: **for** each minibatch of data molecules G_0 **do**
- 2: Simulate legal forward trajectories ω with Algorithm 2; cache states, times, and inverse action metadata.
- 3: Reverse each path to obtain $Y_s = X_{T-s}$.
- 4: Estimate each holding-time integral by quadrature or uniform time samples of λ_θ .
- 5: Add the negative log alias-aggregated rate for every observed reverse transition.
- 6: Add optional cycle-consistency and scheduler-calibration regularizers.
- 7: Update θ ; update prior parameters η only in a separately controlled calibration phase.

D.3 MODEL ARCHITECTURE

Our implementation uses a permutation-equivariant graph Transformer or message-passing network and exposes:

- graph-level embeddings for the total event clock and operator-family scheduler;
- node logits for insert, delete, and substitute operands;
- pair logits for bond and ring operands, masked by adjacency, bridge status, and residual valence;
- parameter logits for atom/charge labels and bond-order changes;
- a time or corruption-state embedding shared across heads.

Raw time may not identify corruption equally across molecules. State-derived features such as size, cycle rank, legal-action counts, and learned corruption embeddings may help, but no feature may rely on the unknown clean source at inference.

D.4 GENERATION

Generation begins from an independently equilibrated prior sample. The exact sampler repeatedly evaluates $R_{\theta,s}$, samples a waiting time, and commits one legal instruction. To reduce terminal mismatch, one may either increase forward horizon until $p_T \approx \pi_0$ or train with terminal states actually produced by the forward chain and quantify the initialization shift using (16).

E CONFLICT-AWARE MULTIPLE ISSUE

Let $\mathcal{C}(G) = \{a_1, \dots, a_m\}$ be candidate instructions with scores w_i . Define read and write sets over atoms, bonds, and global graph metadata. Pairwise conflicts include shared writes, deletion of another instruction’s operand, and incompatible bond updates. Capacity constraints take the form

$$\sum_i z_i \Delta \nu_{iv} \leq c_v(G), \quad z_i \in \{0, 1\}, \quad (28)$$

for each atom v . Connectivity after several deletions is a joint property and must be checked on the proposed final graph or guaranteed by a stronger sufficient condition.

A deterministic maximum-weight scheduler solves

$$\max_{z \in \{0,1\}^m} \sum_i w_i z_i \quad \text{subject to pairwise, capacity, and serializability constraints.} \quad (29)$$

For generative sampling, deterministic maximization can reduce diversity. A practical stochastic alternative samples candidates sequentially without replacement, re-evaluates legality after every provisional instruction, and atomically commits the resulting canonical sequence. We compare this approximation with exact single issue using two-sample tests in addition to endpoint benchmark scores.

F ADDITIONAL EXPERIMENTAL DETAILS AND RESULTS

F.1 MATCHED BASELINE MATRIX

Table 7: Baseline execution sheet. Record provenance so that published numbers are never confused with matched reruns.

Method	Code/commit	Checkpoint or retrained	Training data	Eval samples	Hardware	Notes
BWFlow	TBD	TBD	TBD	TBD	TBD	TBD
GraphBSI	TBD	TBD	TBD	TBD	TBD	TBD
DiGress	TBD	TBD	TBD	TBD	TBD	TBD
GruM	TBD	TBD	TBD	TBD	TBD	TBD
DISCO	TBD	TBD	TBD	TBD	TBD	TBD
Cometh	TBD	TBD	TBD	TBD	TBD	TBD
DeFoG	TBD	TBD	TBD	TBD	TBD	TBD
ConStruct	TBD	TBD	TBD	TBD	TBD	TBD
CoCoGraph	TBD	TBD	TBD	TBD	TBD	TBD
GrIDDD	TBD	TBD	TBD	TBD	TBD	TBD
FragFM	TBD	TBD	TBD	TBD	TBD	fragment-level comparator
MolHIT	TBD	TBD	TBD	TBD	TBD	MOSES comparator
COMPOSE	TBD	retrained	GuacaMol	TBD	TBD	primary

F.2 NOVELTY AND COVERAGE

F.3 TRAJECTORY AND MIXING DIAGNOSTICS

F.4 OPERATION UTILIZATION

For each noise-time bin, plot proposal, legality, MH acceptance, forward execution, and learned reverse execution frequencies for insert, delete, substitute, bond modify, ring close, and ring open.

Table 8: Novelty metrics that are harder to game than exact train-set absence. State the fingerprint and MCS implementation.

Method	Exact novelty $\uparrow$	Scaffold novelty $\uparrow$	NN Tanimoto $\downarrow$	MCS ratio $\downarrow$	Internal diversity $\uparrow$	Duplicates $\downarrow$
DeFoG	TBD	TBD	TBD	TBD	TBD	TBD
CoCoGraph	TBD	TBD	TBD	TBD	TBD	TBD
GrIDDD	TBD	TBD	TBD	TBD	TBD	TBD
COMPOSE	TBD	TBD	TBD	TBD	TBD	TBD

Table 9: Forward and reverse trajectory diagnostics. Report each operator family separately in the supplement.

Variant	Internal valid	RDKit valid	Connected	Accept.	Atom survival	MCS	Terminal AUC
Uncorrected legal walk	TBD	TBD	TBD	–	TBD	TBD	TBD
MH forward, tree initializer	TBD	TBD	TBD	TBD	TBD	TBD	TBD
MH forward, equilibrated prior	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Learned reverse, single issue	TBD	TBD	TBD	–	TBD	TBD	–
Learned reverse, multiple issue	TBD	TBD	TBD	–	TBD	TBD	–

Also report net change in atom count and cycle rank. Nominal support is not enough: if birth/death rates vanish in practice, the model is effectively fixed-size.

F.5 CONDITIONAL EDITING AS A SECONDARY EVALUATION

After the unconditional model is validated, evaluate source-conditioned editing using the same scheduler initialized at a supplied molecule. Tasks may include scaffold-preserving decoration, property optimization under a similarity constraint, and formula-changing optimization. Report success, similarity, edit distance, and all-step validity. Keep this section secondary so the paper’s de novo claim does not depend on conditional use.

F.6 SECONDARY DISTRIBUTION-LEARNING BENCHMARKS

Table 10: Secondary matched evaluation. Use the metric definitions native to each benchmark, but compute every row with one evaluator. NSPDK is included where standard.

Dataset	Method	Valid $\uparrow$	Unique $\uparrow$	Novel $\uparrow$	FCD $\downarrow$	NSPDK $\downarrow$	All-step valid $\uparrow$
MOSES	DeFoG	TBD	TBD	TBD	TBD	TBD	TBD
MOSES	BWFlow	TBD	TBD	TBD	TBD	TBD	TBD
MOSES	GraphBSI	TBD	TBD	TBD	TBD	TBD	–
MOSES	FragFM	TBD	TBD	TBD	TBD	TBD	–
MOSES	MolHIT	TBD	TBD	TBD	TBD	TBD	TBD
MOSES	CoCoGraph	TBD	TBD	TBD	TBD	TBD	TBD
MOSES	GrIDDD	TBD	TBD	TBD	TBD	TBD	TBD
MOSES	COMPOSE	TBD	TBD	TBD	TBD	TBD	TBD
QM9	DiGress	TBD	TBD	TBD	TBD	TBD	TBD
QM9	DeFoG	TBD	TBD	TBD	TBD	TBD	TBD
QM9	BWFlow	TBD	TBD	TBD	TBD	TBD	TBD
QM9	GraphBSI	TBD	TBD	TBD	TBD	TBD	–
QM9	CoCoGraph	TBD	TBD	TBD	TBD	TBD	TBD
QM9	GrIDDD	TBD	TBD	TBD	TBD	TBD	TBD
QM9	COMPOSE	TBD	TBD	TBD	TBD	TBD	TBD

Table 11: Quality–cost reporting. Separate model evaluations from accepted graph edits because multiple issue changes their ratio.

Method	Params	Train GPU-h	Peak GB	NFE/sample	Accepted edits	Samples/s	FCD ↓
DeFoG	TBD	TBD	TBD	TBD	–	TBD	TBD
BWFlow	TBD	TBD	TBD	TBD	–	TBD	TBD
GraphBSI	TBD	TBD	TBD	TBD	–	TBD	TBD
CoCoGraph	TBD	TBD	TBD	TBD	TBD	TBD	TBD
GrIDDD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
COMPOSE, single issue	TBD	TBD	TBD	TBD	TBD	TBD	TBD
COMPOSE-MI	TBD	TBD	TBD	TBD	TBD	TBD	TBD

F.7 COMPUTE, SCALING, AND SAMPLING COST

F.8 CONFIRMATORY RE-EVALUATION OF DEVELOPMENTAL DIAGNOSTICS

Early experiments are used only to formulate hypotheses and choose which ablations to run. The reported values below are confirmatory reruns under the final data split, model budget, training schedule, seed policy, and evaluator; exploratory values are not carried into the submission.

Table 12: Disposition of prior and process diagnostics. Each row becomes scientific evidence only after a frozen confirmatory rerun.

Diagnostic	Hypothesis tested	Final comparison	Status
Single-atom source	Near-autoregressive growth creates one-sided current	Same model, same compute, single-atom prior	TBD
Random tree source	Tree noise creates a ring-installation bottleneck	Tree versus coarse Gibbs prior	TBD
Tree plus random rings	Random closures may create long repair paths	Constructor ablation with equal burn-in	TBD
Uncorrected legal walk	Legal-action degree biases the terminal law	No-MH versus MH with matched proposals	TBD
Flat action head	Large operator families dominate optimization	Flat versus hierarchical scheduler	TBD
Multiple issue	Parallel compatible edits reduce NFE with bounded bias	Exact single issue versus COMPOSE-MI	TBD

F.9 HYPERPARAMETERS AND FROZEN EVALUATION CONTRACT

Table 13: Hyperparameters and evaluation settings. Values are frozen before test-set generation.

Model setting	Value	Evaluation setting	Value
Encoder layers/width	TBD	Generated samples per seed	TBD
Attention heads	TBD	Number of seeds	TBD
Time embedding	TBD	Fingerprint/radius/bits	TBD
Optimizer and LR schedule	TBD	FCD implementation/version	TBD
Batching by graph size	TBD	MCS timeout and normalization	TBD
Forward horizon/schedule	TBD	Bootstrap replicates	TBD
Prior burn-in/thinning	TBD	Hardware and batch size	TBD
Multi-issue candidate budget	TBD	Timeout/failure policy	TBD

F.10 CONDITIONAL EDITING AS A SECONDARY CAPABILITY

For source-conditioned experiments, the initial graph is supplied by the evaluator rather than sampled from π_0 . We report scaffold-preserving decoration, similarity-constrained property optimization, and formula-changing optimization using success rate, property gain, source similarity, edit distance,

all-step validity, and diversity across repeated edits. These experiments demonstrate reuse of the ISA and scheduler but are not used to establish the unconditional de novo claim.

G ETHICS AND BROADER IMPACT

Constraint-preserving generation can improve the reliability and interpretability of molecular design systems, but molecular generation is dual use. Generated structures are hypotheses, not evidence of safety, efficacy, synthesizability, or biological activity; any downstream claim requires independent computational and experimental validation. Release decisions should document benchmark scope, model limitations, intended-use restrictions, and misuse review. The present work uses public molecular datasets and does not involve human subjects or personal data.

H LARGE LANGUAGE MODEL USAGE STATEMENT

A large language model was used to assist with literature discovery, manuscript outlining, mathematical exposition, and language editing. The authors independently verified the cited sources, mathematical statements, implementation, and all empirical results, and take full responsibility for the submission. The model was not treated as an author and was not used to generate or fabricate experimental measurements.